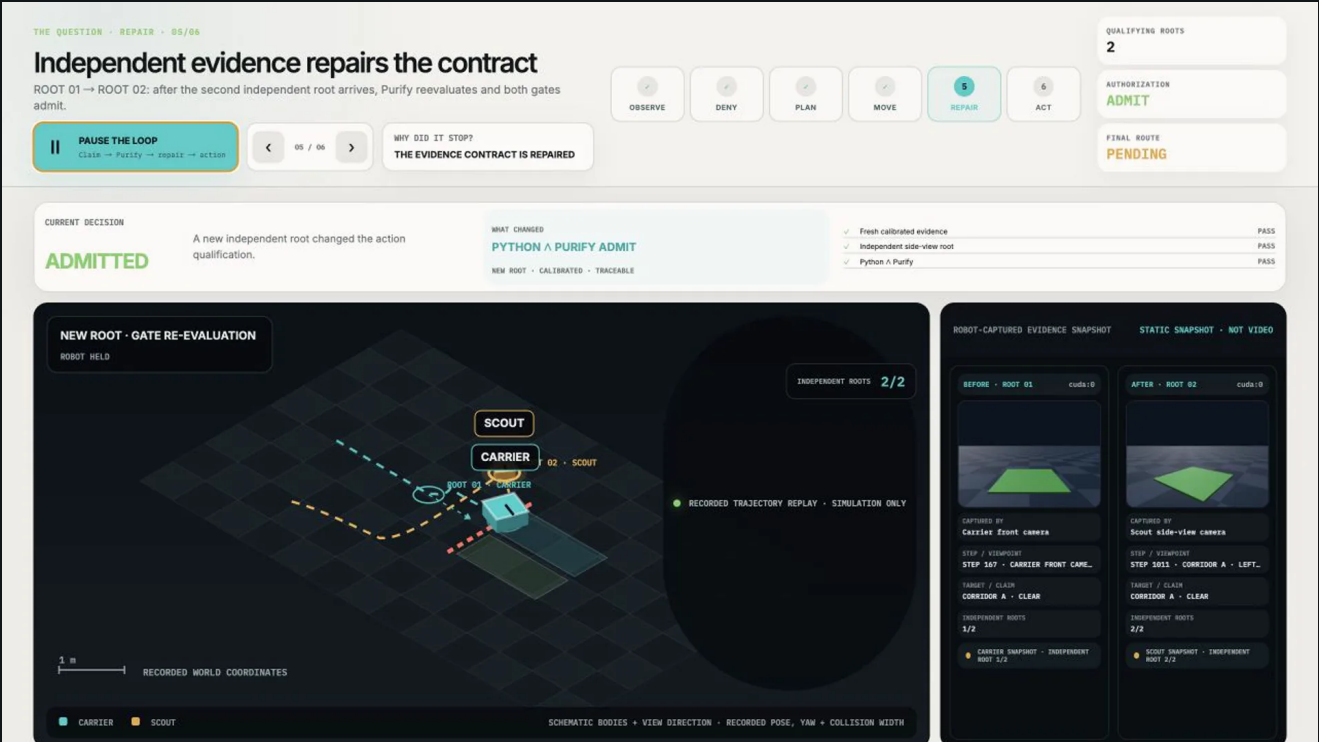


Look Twice V8

Active Evidence Assurance for Physical AI
Technical Report · Frozen Competition Candidate



3,200

LOCKED SAMPLES

29 / 29

FEASIBLE-WORLD DIRECT*

60 / 60

MISSION SUCCESS

0 / 60

UNSAFE OR FALLBACK

* Post-hoc descriptive oracle audit: 29/29 worlds with a clear corridor went direct; the sole dual-blocked world safely detoured.

The preregistered primary remains active 29/30 versus passive 0/30.

Separate additive evidence: 20/20 fixed dual-body dynamics seeds; not a frozen-policy rerun or physical-robot result.

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Executive summary

Robot perception systems normally convert sensor outputs directly into state estimates and actions. This is dangerous when apparently independent outputs are copies of one capture, when observations are stale or time-skewed, when modalities disagree, or when the model is outside its calibration domain.

Look Twice V8 is a pre-action evidence-assurance layer for embodied systems, not another perception leaderboard. It turns Genesis RGB-D observations into spatially grounded, lineage-aware Claims; applies split-conformal prediction sets; evaluates a scoped Action Contract; and requires agreement between the Python control path and a standalone Purify Go gate before a direct physical action is allowed. A denied contract emits a machine-readable BeliefGap. An active policy can then move a scout to an independent viewpoint, acquire a new measurement root, invalidate stale plans, and ask for authorization again. A passive policy denies and takes a disclosed safe detour.

The frozen V8 candidate passed its single-use locked evaluation. The offline population contained 3,200 corridor samples. It achieved 1.000 corridor ROI IoU, 1.000 decisive balanced accuracy, 1.000 decisive blocked recall, 0.000 false-clear singleton rate, and 1.000 conformal coverage. In 12 paired live full-chain seeds, active repair obtained 11/12 direct action qualifications, while the passive policy obtained 0/12 direct routes and detoured 12/12. This is a paired direct-route gain of 91.7 percentage points. Both policies completed 12/12 missions; the 24 policy runs contained zero unsafe crossings, zero unplanned fallbacks, and 24/24 Python/Go decision agreement.

An additive challenge was then bound publicly before execution, without model, calibration, threshold, seed, retry, or stopping-rule changes. Across 30 previously untouched same-generator worlds, active repair achieved 29/30 full-chain direct routes versus 0/30 for passive control, a paired gain of 96.7 percentage points (exact two-sided McNemar $p=3.73e-9$). All 60 episodes completed with zero unsafe crossings and zero fallbacks. The complete 195-file raw result, full subprocess-wall AMD telemetry, checksums, and an independent recomputation validator are public. This supplement is not called locked, OOD, or real-world evidence.

The preregistered primary remains active 29/30 versus passive 0/30. A separate post-hoc descriptive offline oracle-feasibility audit found that all 29/29 worlds with at least one oracle-clear corridor went direct; seed 102515 was the only world with both corridors oracle-blocked and completed safely by detour. Consequently, 30/30 active route outcomes matched offline feasibility. Oracle labels were never available to the controller, and this is not a preregistered endpoint.

The frozen policy result is simulation-only and uses a kinematic Genesis motion backend. A separate submission-time dual-body supplement instantiated 40 non-fixed carrier/scout entities and passed a fixed 20-seed wheel-dynamics bar with zero blocker or pair contacts and zero post-build script pose writes. It is additive, non-locked, and not a rerun or replacement of the frozen policy endpoint.

A second additive decision-bound dynamics replay then consumed the immutable 30 archived challenge outcomes--29 direct corridor decisions and the single dual-blocked safe detour at seed 102515--without rerunning the live policy. It passed 30/30 fixed, serial Genesis/ROCm scenes: all 90/90 distinct non-fixed robot instantiations reached, mean loaded-carrier path fell from 6.2433 m passive to 4.9435 m active (20.8183%), blocker and active-pair contact rows were zero, maximum tilt was 10.5796 degrees, and maximum parked-partner drift was 0.022329 m. The 90 bodies are totals across 30 independent scenes, not one simultaneous 90-body execution. Both dynamics supplements remain non-locked, carry `formal_result_eligible=false`, and claim neither real-robot validation nor safety certification. The frozen primary remains 29/30 direct.

1. Target application

The primary application is a warehouse autonomous mobile robot deciding whether to traverse a designated corridor. A carrier wants to execute the task, while a scout can move to a diagnostic viewpoint when the carrier's front observation is insufficient.

The problem is not only obstacle detection. The system must determine whether the evidence supporting a specific corridor-crossing action is:

- spatially relevant to the corridor under judgment;
- fresh and within an allowed time window;

- produced by known sensor and calibration versions;
- composed of enough independent physical measurement roots;
- free from unresolved RGB, depth, geometry, and map conflicts;
- inside the declared calibration domain.

This assurance pattern is useful beyond warehouses. It can sit between perception and action in inspection robots, delivery robots, autonomous vehicles, and manipulation systems that must act under partial observability without confusing sensor count with evidential independence.

2. System architecture

```
Carrier RGB-D + optional Scout RGB-D
|
v
Corridor projection + spatial RGB-D model
|
v
Robot Claims with scope, time, calibration, capture and device lineage
|
v
Root collapse + split-conformal prediction sets + modality conflict checks
|
v
Scoped Action Contract
|
+--> Python policy admission
|
+--> Purify Go GateReceipt
|
v
effective_admit = python_admit AND go_admit
|
+--> direct route
+--> BeliefGap -> scout observation -> re-evaluate
+--> safe detour or fail closed
```

2.1 Components

Component	Responsibility
Genesis runtime	Simulates the warehouse scene, carrier, scout, cameras, and routes.
Corridor projection	Projects the action-specific corridor into the image as a binary ROI mask.
V8 spatial RGB-D model	Produces ROI-gated obstacle segmentation and a corridor-level blocked probability.
Robot Claim layer	Adds scope, time, calibration identity, source artifact, capture root, and device root.
Conformal layers	Convert model and Go-fusion scores into <code>{clear}</code> , <code>{blocked}</code> , or inconclusive sets.
Action Contract	Defines the evidence conditions required for corridor traversal.
Purify Go core	Evaluates the contract independently and emits canonical hashed receipts.
Active repair planner	Selects a side-view observation that targets the declared BeliefGap.
Evidence Console	Replays recorded evidence without requiring a live GPU.

2.2 Trust boundary

Clean simulator state, clean segmentation, and oracle obstacle labels are available only to dataset construction and evaluation. They are not inputs to the online policy, V8 runtime Claims, repair planner, or Purify gate.

RGB and depth captured at the same time from the same camera count as one physical measurement root. Creating multiple modality Claims does not create multiple independent observations. A second root requires a distinct capture and known device lineage.

3. Dataset and split discipline

The dataset is self-built in Genesis. Each world contains paired views of two candidate corridors. Runtime-legal inputs are RGB, noisy depth, an action-specific corridor mask, and camera/corridor geometry. Clean entity segmentation and clean depth are label-only channels.

3.1 Sample contract

Each sample contains:

- RGB image;
- noisy depth image;
- projected target-corridor mask;
- camera position, yaw, range, and corridor identity;
- clean obstacle mask and corridor state for training/evaluation only;
- world seed, source hashes, and eligibility markers.

3.2 Frozen partitions

Split	Seeds	Worlds	Purpose
Train	100000-101499	1,500	Model fitting only
Validation	101500-101799	300	Checkpoint selection only
Vision calibration	101800-102099	300	Split-conformal vision artifact
Locked test	102100-102499	400	Opened once after the runtime freeze
Preregistered challenge	102500-102529	30	Same generator family; evaluated once after public binding
Remaining challenge reserve	102530-102699	170	Same generator family; not evaluated

The selected checkpoint records 12,000 training pairs and 2,400 validation pairs. The locked report contains 3,200 corridor samples: 1,560 blocked and 1,640 clear.

A separate Go-fusion calibration collection used seeds 105000-105199 and contained 400 corridor rows with both classes represented. Development and confirmatory full-chain smokes used disjoint seeds 105300-105311 and 105400-105411. The locked split was not used by these steps.

The challenge subset changed seeds but not the generator family. Seeds 102500-102529 were evaluated once under the public frozen-challenge protocol; seeds 102530-102699 remain unevaluated. Neither subset is presented as OOD or domain-generalization evidence.

4. V8 perception model

The frozen model is [look-twice-v8-vision/spatial_rgbd_seg_v3/3](#), selected at epoch 22. It contains 39,811,941 trainable parameters. Its state dictionary contains 39,868,705 tensor elements when non-parameter buffers are included.

4.1 Inputs and backbone

- RGB: DeepLabV3 with a ResNet-50 backbone;
- depth: a lightweight convolutional depth encoder;
- spatial scope: a projected binary corridor mask;
- geometry: a 12-dimensional camera/corridor vector;
- input resolution: 256 x 256.

The decoder receives RGB features, depth, and the corridor mask. Obstacle logits are explicitly suppressed outside the target ROI. The traversability head uses masked pooling, so an obstacle in the other corridor cannot dominate the action-specific prediction. The model also produces visibility, quality, and uncertainty values.

4.2 Frozen checkpoint identity

Field	Value
Backend	deeplabv3_resnet50
Epoch	22
Preprocessing	v8-spatial-rgbd-seg-v3-roi-gated/v1
Checkpoint SHA256	7b158726f9c00e01eec7f995674001727be03b84ff684a0cb43cba8682cd5783
Vision conformal identity	ca4a7203eeeedbc0a155955237ffb884f7684a4431e894855f847ee69c5eed1f

The checkpoint was frozen before the locked split opened. The release verifier guards the exact runtime source files and calibration artifacts by SHA.

5. Evidence qualification and active repair

5.1 Robot Claims

A Claim includes the proposed corridor state and the context required to audit it: actor, observer, corridor, action, validity window, calibration identity, capture root, device root, artifact lineage, and supporting values.

5.2 Split-conformal prediction sets

The vision score is converted into one of three operational states:

- singleton {clear};
- singleton {blocked};
- dual or empty set, treated as inconclusive.

Go fusion has its own calibration identity. This prevents a Python-side score from silently bypassing the independently evaluated action contract.

5.3 Purify Action Contract

Direct traversal is admitted only when the evidence meets the declared scope, freshness, root, conflict, prediction-set, and calibration conditions. Purify returns a deterministic GateReceipt that lists accepted and discounted Claims, root identities, failed clauses, BeliefGaps, and a canonical SHA256.

The final action signal is:

```
effective_admit = python_admit AND purify_go_admit
```

A model cannot grant action authority by itself.

5.4 Active versus passive behavior

The passive policy stops direct traversal after the initial denial and follows a safe detour. The active policy uses the BeliefGap to select an independent side-view capture. It then rebuilds the Claims and asks both authorization paths again. This separates safety from usefulness: denial remains safe, while active repair can recover a shorter direct route when the evidence supports it.

The intended operational trade is specific: move a low-risk scout to repair evidence so a loaded carrier can avoid a conservative detour. It is not a claim that active repair reduces total logical-role team travel or wall-clock time.

6. AMD Radeon GPU and ROCm use

The frozen evaluation ran on one Radeon Cloud `gfx1100` GPU with approximately 48 GiB VRAM.

Layer	Execution
Genesis simulation	Radeon GPU through <code>gs.amdgpu</code>
RGB-D rendering	Radeon GPU
RGB-D preprocessing	PyTorch ROCm tensors
Spatial model inference	Radeon GPU through PyTorch ROCm <code>cuda:0</code>
Purify contract gate	CPU, standalone Go process
Replay website	CPU-only recorded-evidence presentation

The recorded software stack was Genesis 1.1.2, PyTorch 2.9.1+gitff65f5b, and HIP 7.2.53211-e1a6bc5663 on Linux. `cuda:0` is the PyTorch ROCm device namespace and does not indicate an NVIDIA execution path.

6.1 Frozen-model inference benchmark

A submission-preparation benchmark loaded the exact checkpoint above after verifying its SHA256. It used preloaded 256 x 256 FP32 synthetic tensors, 20 warm-up iterations, and 100 synchronized model forwards per batch.

Batch	p50	p95	Throughput	Peak allocated
1	192.31 ms	199.18 ms	5.18 images/s	313.63 MiB
4	659.08 ms	667.85 ms	6.06 images/s	465.38 MiB
8	1,279.72 ms	1,288.29 ms	6.25 images/s	668.38 MiB

These numbers include the vision forward pass, Python dispatch, and synchronized ROCm execution. They exclude RGB-D preprocessing, Genesis, Go fusion, I/O, and robot actuation, so they are not end-to-end loop latency. No GPU-utilization claim is made. The machine-readable report is [release/v8-frozen/results/V8_FROZEN_INFERENCE_BENCHMARK.json](#), with SHA256 `282b0a1bf5180d9aca75cb068b60100222eb07bf6aea9fc8a2ca46c655b14156`.

6.2 Sustained ROCm telemetry

An additive 60-second batch-8 run used the same checkpoint, source hashes, preloaded FP32 tensors, and synchronized forward scope. Its hard preflight recorded 0% GPU use, 0% VRAM allocation, and no KFD process before model load. Across 60.182 measured seconds, 61/61 `rocm-smi` samples reported 100% GPU use; mean graphics-package power was 135.33 W, p95 power was 156 W, and 376 images were processed at 6.248 images/s.

This supports a sustained Radeon execution claim only. It is not an accuracy run or locked-test rerun and excludes Genesis, preprocessing, Go fusion, I/O, actuation, mission energy, and end-to-end robot latency. The raw samples

are in [release/v8-frozen/results/V8_FROZEN_ROCM_TELEMETRY.json](#), SHA256
 0ec12a92ac4e88a97d9068e40a06f72f9dd5ecaa16503c45e2d965d4d876dde9.

6.3 Frozen-challenge full-pipeline telemetry

Unlike the isolated throughput runs above, the 30-world challenge retained two-second `rocm-smi` samples across the complete Python, Genesis, checkpoint, and episode-subprocess wall. All 60 episodes used live Genesis RGB-D, loaded the frozen checkpoint, and emitted Purify Go receipts. The run produced 268 RGB-D observations, 134 vision proposals, and 268 Go invocations and receipts.

Metric	Full-wall result
Episodes	60 / 60 valid
Measured subprocess wall	1,685.5 s
ROCM samples	844 at 2 s intervals
GPU use, mean / median / p95 / max	19.4 / 0 / 95 / 100%
VRAM allocation, p95 / max	2 / 2%
Graphics package power, mean / p95 / max	35.7 / 81 / 109 W
Active subprocess wall, mean / p95	29.92 / 31.16 s
Passive subprocess wall, mean / p95	26.26 / 26.70 s

Zero-percent idle samples are deliberately included. These measurements show that the recorded challenge used the full Radeon pipeline; they are not control-loop latency, mission energy, or hardware-utilization optimization claims. The retained telemetry is [release/v8-frozen/results/challenge_102500_102529/ROCM_TELEMETRY.json](#).

6.4 Additive dual-body rigid-dynamics execution

The frozen V8 policy realizes carrier and scout as distinct logical poses on one shared kinematic chassis. To test the narrower actuation gap separately, a fixed submission-time protocol instantiated one loaded carrier and one scout as distinct non-fixed Genesis URDF bodies for each of 20 seeds. All 40 robot entities shared one non-overlapping scene. After `scene.build()`, the script used only wheel-DOF `control_dofs_velocity`; it issued no entity pose writes.

The complete run used Genesis 1.1.2 on [gs.amdgpu](#), PyTorch 2.9.1+gitff65f5b, and HIP 7.2.53211-e1a6bc5663. It ran for 4,299.992 seconds and passed 20/20 fixed seeds. This is an acceptance execution, not a throughput, energy, or control-loop-latency benchmark.

The first whole-process attempt reached an external 3,600-second watchdog before the script wrote its single end-of-run report; no per-seed outcome was observed. Recovery increased only that external watchdog to 10,800 seconds. Source commit, runner and URDF bytes, seeds, protocol values, and thresholds were unchanged. Both attempt records are retained.

6.5 Additive decision-bound rigid-dynamics execution

The 20-seed supplement above establishes a component-level carrier/scout wheel-dynamics bar. A separate recovery V2 closes the narrower archived decision-to-actuation boundary for all 30 fixed challenge worlds. Each seed ran serially in a fresh [gs.amdgpu](#) subprocess containing one active scout, one active loaded carrier, and one passive loaded carrier. Active and passive replicas remained paired within the same scene; unrelated cross-seed bodies did not share a scene. Across the complete denominator, this instantiated 90 distinct non-fixed robot bodies and sealed 30 per-seed checkpoints.

The Radeon environment was Genesis 1.1.2, PyTorch 2.9.1+gitff65f5b, and HIP 7.2.53211-e1a6bc5663. All 30 workers completed on attempt 1 with exit code zero. The retained approximately 18-minute wall is an execution-audit fact, not throughput, latency, utilization, or energy evidence. The replay used the archived route decision for each world; it did not rerun the frozen RGB-D perception-policy loop.

7. Evaluation protocol

7.1 Freeze and open discipline

Before the locked split was opened, the checkpoint, two conformal identities, Purify binary, runtime files, seed ranges, and promotion gates were frozen. The locked open produced a separate seal. The report is permanent and declares:

- `no_retune=true`;
- `no_refit=true`;
- `no_vision_retrain=true`;
- no errors.

7.2 Offline metrics

The offline evaluator measures corridor ROI IoU, decisive balanced accuracy, blocked recall, false-clear singleton rate, conformal coverage, and the number of decisive prediction sets.

7.3 Live full-chain metrics

For each of 12 locked seeds, the passive and active policies run in the same scenario family. The report checks initial denial, active repair, final direct qualification, passive detour, unsafe crossing, unplanned fallback, Purify invocation, and policy-shape invariants.

7.4 Publicly preregistered frozen challenge

Before any challenge episode ran, Commit A published the protocol, runner, validator, seed range 102500-102529, metrics, stopping rules, and failure handling. Commit B then bound the exact Commit A hash and changed only that field. The public timestamps were 2026-08-03 10:01:53Z and 10:02:22Z.

The runner attempted each of 30 worlds exactly once for each policy, retained timeouts and nonzero exits as results, and prohibited retry, replacement seed, or early stopping. The primary endpoint was deliberately stricter than route choice: a full-chain direct outcome required mission success, direct routing without detour, a nonempty executed corridor, and a same-corridor receipt on which Python, Purify Go, and the effective gate all admitted. A separate validator read the 60 raw episode records, recomputed the report without loading the checkpoint or running an episode, and returned `passed=true` with zero errors.

8. Results

8.1 Locked offline population

Metric	Result
Samples	3,200
Blocked / clear	1,560 / 1,640
Corridor ROI IoU	1.000
Decisive balanced accuracy	1.000
Decisive blocked recall	1.000
False-clear singleton rate	0.000
Conformal coverage	1.000
Decisive samples	3,001 / 3,200

8.2 Locked live full chain

Metric	Result
Active effective admissions	11 / 12
Active full-chain direct routes	11 / 12
Passive full-chain direct routes	0 / 12
Paired direct-route gain	+91.7 percentage points
Mission completion	12 / 12 active; 12 / 12 passive
Passive initial denials	12 / 12
Passive safe detours	12 / 12
Passive repair attempts	0 / 12
Unsafe crossings	0 / 24 policy runs
Unplanned fallbacks	0 / 24 policy runs
Python / Purify Go decision agreement	24 / 24 policy runs

One active seed remained conservative and detoured rather than obtaining final direct qualification. Seed 102105 is counted in the denominator and not removed. Among the 11 discordant direct-route pairs, all 11 favored active repair. An exact two-sided McNemar test gives $p=0.0009766$; this is descriptive evidence for the fixed seed suite, not a population or real-world guarantee.

8.3 Preregistered 30-world frozen challenge

Metric	Result
Valid paired worlds / episodes	30 / 60
Active full-chain direct routes	29 / 30 (96.7%)
Active Wilson 95% interval	83.3-99.4%
Passive full-chain direct routes	0 / 30
Passive Wilson 95% interval	0-11.4%
Paired direct-route gain	+96.7 percentage points
Exact two-sided McNemar	$p=3.73e-9$
Mission completion	60 / 60
Unsafe crossings / fallbacks	0 / 60; 0 / 60
Live RGB-D / checkpoint / Go receipt	60 / 60 each
Receipt-level Python/Go agreement	250 / 268 (93.3%)

Formal primary stays active 29/30 versus passive 0/30 (exact McNemar $p=3.73e-9$). A descriptive post-hoc offline oracle audit - not available to the controller or preregistered - found 29/29 clear-corridor worlds direct via a clear corridor; dual-blocked seed 102515 safely detoured. Thus 30/30 route outcomes matched offline feasibility without relabeling; all 30 active records had zero unsafe crossings, collisions, and fallbacks.

All 18 receipt disagreements were Python-admit/Go-deny with `effective_admit=false`; none opened the gate. Every disagreement was then examined in a post-hoc descriptive audit, not a preregistered endpoint. All shared the same signature: active policy, corridor B, only one Go-qualified measurement root, and the inconclusive Go set `{clear, blocked}`. Four concerned a corridor not finally selected; 14 were transient evaluations before a later

selected-corridor joint admit. The audit changed no runtime, calibration, or threshold. Each of the 29 active direct successes had a same-corridor receipt on which Python, Go, and the effective authorization were jointly true.

The operational burden moved in the intended warehouse direction but was not free. Mean loaded-carrier logical path fell from 6.404 to 4.961 (-22.5%), while the active scout added 2.980. Mean total logical-role team path therefore rose from 6.404 to 7.941 (+24.0%). These are kinematic logical-role paths on one shared Genesis chassis, not energy, throughput, latency, physical duty cycle, or simultaneous two-robot dynamics.

8.4 Guarded confirmatory cost ledger

The non-locked seed 105400 replay includes trajectory lengths and is guarded by the frozen import manifest. It is kinematic simulation and carries `formal_result_eligible=false`; it is not added to the locked aggregate.

Metric	Active repair	Passive baseline
Loaded-carrier travel	4.915 m	6.404 m
Scout travel	3.046 m	0.000 m
Total robot travel	7.961 m	6.404 m
Episode steps	1,984	1,150
Observations / replans	3 / 2	1 / 0
Payload delivered / collisions	yes / 0	yes / 0

Active repair reduced loaded-carrier travel by 1.488 m, or 23.24%, but raised total robot travel by 24.32%. This supports a burden-shifting claim - from the loaded carrier to a diagnostic scout - not a total-distance or latency speedup. The derivation is machine-readable at [release/v8-derived/V8_TASK_UTILITY_DERIVATION.json](#), SHA256 `f85f6d647ea49f9bc148cf9fad6c38a34050cd8e9f8f690522b965c5ff23730b`.

8.5 Additive 20-seed dual-body rigid dynamics

Fixed check	Result
Confirmatory seeds	20 / 20 passed
Distinct non-fixed robot entities	40
Carrier / scout reached	20 / 20; 20 / 20
Trial-blocker / carrier-scout contact rows	0 / 0
Script entity pose writes after build	0
Maximum body tilt	10.750 degrees (limit 20)
Maximum parked-partner drift	0.018061 m (limit 0.08)
Minimum scout / carrier path	0.940847 / 4.746561 m
Maximum scout / carrier goal error	0.139779 / 0.139922 m (limit 0.14)

The goal-error maximum sits close to the fixed stopping tolerance because the controller stops on first entry into the 0.14 m goal region; it is not presented as a large-margin result. All stability, parked-drift, nontrivial path, contact, and source-boundary checks passed. The same report was accepted by the verifier on the Radeon host and locally.

The report has SHA256 `8a883163ff544bdf7aa9410b4b4d364e88dcee15dce15edcbd791a1d4b4fd110`. It establishes bounded sequential wheel motion by separate carrier and scout rigid bodies. It does not establish a full-policy conversion to simultaneous two-robot control, physical-robot transfer, dynamic-obstacle response, energy savings, or a new formal V8 endpoint.

8.6 Additive 30-seed decision-bound rigid dynamics

Fixed check	Result
Fixed scenes / sealed checkpoints	30 / 30 passed; 30 / 30 sealed
Archived decision mix	29 direct + 1 safe detour (102515)
Distinct non-fixed robot instantiations	90 across 30 serial scenes
Scout / active carrier / passive carrier reached	30 / 30 / 30
Mean active / passive loaded-carrier path	4.943529 / 6.243270 m
Paired mean loaded-carrier path reduction	20.8183% (floor 15%)
Direct pairs saving at least 0.50 m	29 / 29
Minimum direct-pair carrier saving	1.333621 m
Blocker / active carrier-scout contact rows	0 / 0
Maximum body tilt	10.579607 degrees (limit 20)
Maximum parked-partner drift	0.022329 m (limit 0.08)
Maximum goal error	0.139930 m (limit 0.14)
Script entity pose writes after build	0

The controller stops on first entry into the fixed 0.14 m goal region, so the goal-error result is not presented as a large-margin endpoint. Every other predeclared per-trial and aggregate check also passed. This result binds the 29 direct choices and the one safe-detour choice to wheel-actuated execution; it does not replace or upgrade the frozen 29/30 direct endpoint.

The report SHA256 is [1501e31bdc1bc353d56224f76f0a0f58de574e6c436980bc9c22a7c33104bd99](#). It is archived-decision simulation evidence across independent serial scenes, not a live-policy rerun, simultaneous cooperative execution, or one simultaneous 90-body scene. It is additive, non-locked, and explicitly records `formal_result_eligible=false`.

8.7 Evidence identity

The authoritative locked report file SHA256 is [5b88d5e7683f853380f1e23123f830c6966824e3afe055af5c4fb6604f672cb](#). The open-seal SHA256 is [7bfe13d0d7e117f76286cb094711322b810dc44d40ddb149bf1304713baba14](#).

An additive pre-open input-and-label archive preserves the exact 400-world, 3,200-record locked population. A streaming verifier checked its 1,019,307,579 bytes, SHA256

[0933053f28aca5254f13eb2eb11ce16c2f488e1880e4e282b1e4dfd7d957cfba](#), all 22,800 file members, seed and label counts, path hygiene, sidecar chronology, and permanent-result identities without extracting arrays or running the model. This pack is input-only: it does not contain the original 3,200 one-shot prediction rows or 24 raw full-chain episodes and therefore cannot recompute the permanent aggregate. Those missing outputs were not regenerated.

The additive challenge raw archive contains 195 files and has SHA256

[171c9bab73554e1a3654c24872ade011b8423d3aca0df8eca38625a90b0854d2](#). Its internal [SHA256SUMS](#) covers every other regular file (194/194). The independent validation report has SHA256

[942f1624e6903033335e5ffbcdbc12afed4a0e8eed4e0d33ffa657f5e147a940](#) and reproduces the compact challenge report exactly from raw episode records.

The decision-bound V2 evidence directory is [release/v8-derived/decision_dynamics_recovery_v2_102500_102529](#). Its complete report SHA256 is [1501e31bdc1bc353d56224f76f0a0f58de574e6c436980bc9c22a7c33104bd99](#); the source-binding SHA256 is [c40f6ba74a39ad761e4926ddb66326f33a1a645fef3c96364f9c53b9d5d3eb5d](#). The 79-entry post-run package checksum index has identity

`PACKAGE_SHA=24d3538365d2818f5e5b64c5f06ecce94bdae1e4d3df2ab320178248bf71540`.

Two proof-scope limits are preserved rather than hidden. The inner V2-run [SHA256SUMS](#) covered the source binding, 30 checkpoints, and report (32 files), but not [ATTEMPTS.json1](#), [PROGRESS.json](#), or worker logs. The retained attempt ledger and logs show exactly 30 attempt-1 completions with exit code zero, but the inner checksum and V2 attempt verifier alone are not a cryptographic proof that no additional unsealed attempt ever existed. The later package index fixes the identities of the retained 79-file package without retroactively expanding that original proof.

Also, the V2-run source binding omitted the directly imported [src/v4_motion.py](#). Its post-run working-tree and commit-[b0c4f0d](#) bytes both hash to [fd94a2d7b89cc3118e3ccf4ae2545f95c84e6596b48c7e79bf52f352935d0772](#), and a clean 2,419-file post-run audit found no source difference or writable non-Git file. That is strong corroborating evidence, not signed continuous source attestation. [PROVENANCE_REVIEW.json](#) records both limits and the safe claim language.

9. Innovation and technical contributions

1. Lineage-aware evidence accounting. The system counts physical roots, not the number of derived model outputs.
2. Spatially scoped perception. The V8 model predicts the state of the corridor named by the intended action instead of asking whether any obstacle exists anywhere in the frame.
3. Conformal action qualification. Uncertainty becomes an explicit prediction set and contract clause rather than an uncalibrated confidence threshold.
4. BeliefGap-driven evidence repair. A denial explains what evidence is missing and selects a physical observation intended to repair that gap.
5. Dual authorization with receipts. The Python controller and standalone Go core must agree, and the decision is retained as a canonical receipt.
6. Replayable proof surface. Judges can inspect a recorded evidence chain and source JSON without needing the live competition GPU.
7. Explicit task-cost accounting. The submission distinguishes loaded carrier burden from total team motion, so active repair is not presented as a free speed or energy improvement.
8. Public one-shot challenge binding. A before/after commit pair fixes the protocol and code before evaluation, while a checksum-complete raw archive supports independent recomputation of every reported challenge endpoint.
9. Explicit dual-body actuation audit. A separate predeclared 20-seed bar instantiates carrier and scout as non-fixed rigid bodies, binds wheel-only post-build actuation and fail-closed stability/contact checks, and retains a transparent infrastructure-timeout recovery chain.
10. Checkpointed decision-to-actuation bridge. A second predeclared layout binds every archived 30-world decision to a fresh paired rigid-body scene, atomically seals the complete fixed-order denominator, and keeps recovery observability separate from the frozen policy endpoint.

10. Real-world value

Industrial robots frequently combine outputs from multiple models trained on the same camera, reused maps, asynchronously delivered messages, and sensor proxies with different calibration domains. Treating those outputs as independent votes can create false certainty.

Look Twice offers a deployable architectural boundary between perception and action. It can reduce unsafe action from correlated evidence while avoiding a robot that simply stops forever: the active observation loop attempts to earn the missing evidence. Receipts provide a useful audit trail for operations, incident review, and future assurance tooling.

11. Deliverables

- dedicated Apache-2.0 source repository;
- public Robot Claim, Action Contract, calibration, receipt, and episode schemas;

- standalone Purify Robotics Go reference core;
- frozen V8 result import and SHA verifier;
- CPU-only Docker Evidence Console;
- source-linked replay bundles and 30-second evidence reel;
- English technical report and detailed reproduction guide;
- a release-hosted pre-open locked input-and-label pack with streaming verifier;
- exact-checkpoint sustained ROCm utilization and power telemetry;
- a publicly preregistered 30-world same-generator challenge with 60 raw episodes, full-wall ROCm telemetry, recursive checksums, and an independent validator;
- a separately verified 20-seed dual-body rigid-dynamics report with 40 non-fixed entities, timeout/recovery audits, and sealed checksums;
- a separately verified 30-seed decision-bound rigid-dynamics replay with 90 distinct non-fixed instantiations, 30 atomic checkpoints, package-wide checksums, and a disclosed post-run provenance-scope review;
- a one-page English Frozen Challenge Judge Card;
- final 3:59 English workflow video with fixed-composition visuals and AI-generated OpenAI Cedar narration;
- public 159,592,901-byte frozen checkpoint release asset.

Public Evidence Console: <https://eason4kim-rocket.github.io/>

Frozen source branch: <https://github.com/eason4kim-rocket/look-twice/tree/v8-competition-release>

Final workflow video: <https://github.com/eason4kim-rocket/look-twice/releases/download/v8-competition-candidate/Look-Twice-V8-Demo.mp4>

Frozen checkpoint: https://github.com/eason4kim-rocket/look-twice/releases/download/v8-competition-candidate/v8_seq_v3_selected_ep22_7b158726f9c0.pt

12. Reproducibility

The fastest audit requires Python 3.11+:

```
python3 scripts/build_competition_replays.py
python3 -m unittest tests.test_competition_replay -v
python3 scripts/verify_frozen_foundation.py
python3 scripts/derive_v8_task_utility.py
```

The separately downloaded locked-input pack can be checked without inference:

```
python3 scripts/verify_v8_locked_input_pack.py \
--archive /path/to/locked.tar.gz --sidecar /path/to/locked.sidecar.json \
--check-only
```

The preregistered challenge can be recomputed from its complete raw archive:

```
curl -L0 https://github.com/eason4kim-rocket/look-twice/releases/download/\
v8-competition-candidate/v8-frozen-challenge-102500-102529.raw.tar.gz
tar -xzf v8-frozen-challenge-102500-102529.raw.tar.gz
python3 scripts/verify_v8_frozen_challenge.py \
--results-dir v8-frozen-challenge-102500-102529 \
--preregistration release/v8-frozen/results/\
V8_FROZEN_CHALLENGE_PREREGISTRATION.json \
--repo-root . \
--output v8-frozen-challenge-102500-102529.LOCAL-VERIFICATION.json
```

The evidence website can be rebuilt with:

```
docker compose up --build
```

The additive dual-body report can be checked without a GPU:

```
DYN=release/v8-derived/dual_body_dynamics_160820_160839
(cd "$DYN" && shasum -a 256 -c SHA256SUMS)
python3 scripts/verify_v8_additive_dual_body_dynamics.py "$DYN/REPORT.json"
```

The additive decision-bound dynamics package can also be checked without a GPU. Verify the package index itself before trusting its 79 entries:

```
V2=release/v8-derived/decision_dynamics_recovery_v2_102500_102529
PACKAGE_SHA=24d3538365d2818f5e5b64c5f06ecee94bdeae1e4d3df2ab320178248bf71540
printf '%s %s\n' "$PACKAGE_SHA" "$V2/PACKAGE_SHA256SUMS" | shasum -a 256 -c -
(cd "$V2" && shasum -a 256 -c PACKAGE_SHA256SUMS)
python3 scripts/verify_v8_additive_decision_dynamics_recovery_v2.py \
"$V2/REPORT.json"
```

Expected verifier output ends with 30/30 and report SHA256

1501e31bdc1bc353d56224f76f0a0f58de574e6c436980bc9c22a7c33104bd99.

The standalone contract core is tested with:

```
cd purify_robotics
go test ./...
```

The full procedure, artifact layout, GPU command, and expected outputs are in [docs/V8_REPRODUCTION.md](#).

13. Limitations and non-claims

- Simulation only; no real-robot or sim-to-real result.
- Kinematic Genesis motion is used in the public V8 evidence path.
- Entity segmentation is a disclosed simulation/evaluation proxy, not a real-world semantic sensor.
- Statistical claims apply only to the declared simulated distributions.
- The public replay is a non-locked confirmatory example, not the locked aggregate.
- The 30-world challenge is an additive publicly preregistered same-generator supplement, not part of the permanent locked result and not an OOD claim.
- Carrier and scout in the frozen policy are separate logical-role poses, viewpoints, and capture roots on one shared Genesis chassis. A separate 20-seed supplement demonstrates bounded sequential wheel motion by two non-fixed rigid bodies, but not a full-policy conversion, simultaneous cooperative control, or two physical devices.
- The 30-seed decision-bound dynamics replay consumes archived route outcomes; it does not rerun live perception or policy inference. Its 90 bodies are distinct instantiations across 30 independent serial scenes, not one simultaneous 90-body scene. It is additive, non-locked, `formal_result_eligible=false`, and is not dynamic-obstacle, real-robot, sim-to-real, energy, throughput, or safety-certification evidence.
- The V2 retained ledger shows 30 first-attempt worker completions and no replacement. Its inner V2-run checksum/attempt-verifier scope did not cover every attempt, progress, and log byte, so this observation is not promoted into a cryptographic proof of the absence of any unsealed attempt.
- The V2-run source binding omitted the direct `src/v4_motion.py` import. Clean post-run tree and commit comparisons found byte identity and no source difference, but they are corroboration rather than continuous attestation.
- Receipt-level agreement in the challenge is 250/268 (93.3%). All 18 disagreements were vetoed by Go and remained fail-closed.
- The locked-input supplement contains inputs and labels, not the original per-sample predictions or 24 raw full-chain episodes; it does not reconstruct the one-shot run.
- Seeds 102500-102529 were evaluated once under the public challenge protocol; seeds 102530-102699 remain unevaluated. Neither is claimed as OOD evidence.
- The frozen 159 MB checkpoint is distributed as a GitHub release asset because it exceeds GitHub's normal 100 MB blob limit; its SHA must be verified before use.
- The source tree was identity-captured by file hashes because the GPU worktree was dirty; the Git commit alone is not presented as the runtime identity.
- The public Purify core is a contest reference implementation, not the private Purify product or a compatibility commitment.

- Later Integrity Shield R1/R2 and V9 research is not promoted into V8 claims.
- A focused Genesis URDF inertial-origin parser fix and required regression test are prepared locally with baseline-fail/patch-pass evidence. No public external upstream contribution is claimed until owner approval.

14. Team and contributions

Liu Liang - solo entrant. GitHub: [@eason4kim-rocket](#)

Project conception, simulation and robot loop, dataset protocol, V8 perception model, conformal calibration, Purify Go reference core, experiment execution, evidence integrity, website, documentation, and submission packaging.

References

1. [AMD AI DevMaster Hackathon official event page](#)
2. [Official submission repository](#)
3. [Genesis physics simulation framework](#)
4. [AMD ROCm documentation](#)